# ARDUINO ROBOTICS & ELECTRONICS

Lesson 1: Mastering LCD Displays & I2C Communication

## 1. What is an LCD Display?

**LCD** stands for **Liquid Crystal Display** . It is an electronic display module used to show text, numbers, and characters .

The standard 16x2 LCD display features **2 rows** and **16 characters per row**, allowing you to display up to **32 characters** at once .

> **Real-Life Examples Around Us:** LCD displays are everywhere! You can see them in railway departure/arrival boards at Indian Railway stations, token counters in banks and hospitals, traffic signals, and digital calculators .

## 2. Wiring Made Easy with I2C Backpack

### Traditional LCD Wiring (16 Wires)

Standard LCDs require connecting up to 16 separate pins to the Arduino . This creates a messy clutter of wires, takes up all digital pins, and easily leads to mistakes !

### I2C LCD Wiring (Only 4 Wires!)

Using an I2C backpack module, wiring becomes super simple and safe—requiring only 4 total wires :

- **VCC → 5V** (Power)
- **GND → GND** (Ground)
- **SDA → Pin A4** (Data)
- **SCL → Pin A5** (Clock)

## 3. Step-by-Step Circuit Assembly & Programming

**Step 1: Power Connections** — Connect VCC to 5V and GND to GND on the Arduino board . Always connect your power lines first !

**Step 2: Data Connections** — Connect SDA to analog pin A4 and SCL to analog pin A5 . Double-check all four connections before powering up .

**Step 3: Writing the Code** — Include the necessary libraries to establish communication :

```
#include <Wire.h> // Helps Arduino communicate over I2C
#include <LiquidCrystal_I2C.h> // Controls the I2C LCD

LiquidCrystal_I2C lcd(0x27, 16, 2);

void setup() {
  lcd.begin(); // Initializes the LCD
  lcd.backlight(); // Turns on the screen backlight
  lcd.setCursor(0, 0); // Set position to Row 0, Col 0
```

```
  lcd.print("Hello, India!"); // Prints message
}
```

## 4. Mini-Project: Smart Toll Booth Counter

**Hands-On Application**

Just like actual toll gates on Indian highways, you can build a **Smart Toll Booth Counter** using Arduino and an I2C LCD display !

- **How it works:** Every time a vehicle passes through the toll sensor, the system increments the tally by 1 and updates the count on the LCD screen in real time .
- **Setup:** Keep the 4-wire I2C setup (VCC→5V, GND→GND, SDA→A4, SCL→A5) , upload your counter sketch, and watch the tally update as vehicles cross !

## 5. Clean-Up & Safety Tips

- **Power Off First!** Always disconnect the USB cable / power source before making or changing any pin connections .
- **Stay Organized:** Keep your workspace tidy and component wires untangled—a clear desk means a clear mind !
- **Handle with Care:** Electronics and LCD glass screens are delicate; handle modules gently without pressing hard on the display .
- **Teacher Supervision:** Always ask your teacher or instructor to inspect your circuit before plugging in the power .

# Practice Questions: Test Your Knowledge

Complete these practice questions in your notebook to test your understanding of LCD displays and I2C wiring .

### Question 1 (Multiple Choice)

What does the abbreviation **LCD** stand for?
A) Light Crystal Diode
B) Liquid Crystal Display
C) Linear Circuit Device
D) Laser Control Display

### Question 2 (Short Answer)

How many total characters can a standard 16x2 LCD display show on its screen at one time? Explain how you calculated this number .

### Question 3 (Wiring Comparison)

Compare traditional LCD wiring with I2C LCD wiring. How many wires does each method require, and why is the I2C method safer and easier for students ?

### Question 4 (Pin Identification)

Match the 4 pins of an I2C LCD module to their corresponding pins on an Arduino Uno board :
1. VCC → ______________
2. GND → ______________
3. SDA → ______________
4. SCL → ______________

### Question 5 (Arduino Code)

Which built-in library helps the Arduino board talk to the LCD using the SDA and SCL data lines over I2C communication ? Write the line of code used to include this library .

### Question 6 (Application & Mini-Project)

Describe how an I2C LCD screen is utilized in the "Smart Toll Booth Counter" mini-project. What information is shown on the display as vehicles cross the booth ?

### Question 7 (Safety Protocols)

State the single most important safety step you must perform before altering any wire or component in an Arduino circuit .